PlatePulse – Hackathon Scope Document

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1. Hypothesis Alignment

Problem Hypothesis

Urban Indian gym-goers (18–35) fail to achieve fitness goals because logging Indian meals is inaccurate, time-consuming, and guess-based, leading to low confidence and app abandonment.

Customer Hypothesis

Primary user:

- **Urban Indian gym-goer**.
- Eats home-style Indian food (dal, sabzi, rice, roti)
- Cares about macros (protein/fat/carbs)
- Wants fast, accurate logging without manual input

Solution Hypothesis

If users can **snap a photo of their Indian meal and instantly receive gram-weight and macro estimates**, then:

- Logging friction reduces
- Confidence increases
- Retention improves

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2. Scope Boundary

We will build a **functional clickable prototype** with:

- Image upload / camera capture
- Backend call (real model OR dummy API stub)

- Display:
  - Segmented food items
  - Estimated weight (grams)
  - Calories
  - Macro breakdown
- Log meal confirmation
- Simple meal history screen

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3. Technical Assumptions

Based on the Food Scanner pipeline

Pipeline components:

- Zero-shot segmentation
- Prototype classification
- Weight regression head (ResNet-50)
- Nutrition computation

For hackathon feasibility:

- Use pretrained checkpoints (if available)
- Run Food Scanner FastAPI demo

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4. Top 3 Use Cases

Use Case 1 – Snap & Analyze Meal

User Goal

Log lunch in under 30 seconds.

User Flow

1. Home Screen → Tap "Snap Meal"
2. Camera / Upload Image
3. Loading screen ("Analyzing your plate…")

4. Results screen:
    - Detected items
    - Weight in grams
    - Calories
    - Macros (Protein / Carbs / Fat)

Backend Flow

Image → API → JSON response → Render results

Success Criteria

- User gets structured breakdown
- No manual typing required

Use Case 2 – Confirm & Log Meal

User Goal

Add analyzed meal to daily log.

User Flow

1. Results screen
2. Tap "Confirm & Log"
3. Success screen → “Added to Today”

Backend

- Store JSON in temporary database (or in-memory list)

Success Criteria

- Meal stored
- Totals updated

Use Case 3 – View Daily Nutrition Summary

User Goal

Check if protein target is met.

User Flow

1. Home Screen → "Today's Summary"
2. Display:
    - Total calories
    - Total protein
    - Total carbs
    - Total fat
    - Progress bar toward daily goal

Backend

Aggregate logged meals

Success Criteria

- Totals update dynamically
- Clear visual macro distribution

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5. System Architecture (MVP)

Frontend:

- React / Flutter / Simple Web App

Backend:

- FastAPI

Modules:

- /analyze → returns food breakdown
- /log → store meal
- /summary → return aggregated totals

Note: Model training may not be done by the time the hackathon ends.